Does Adaptive Risk Aversion Pay? A Pre-Registered Evaluation of Probabilistic Day-Ahead Drawal Scheduling Under India's Deviation Settlement

Rohit Kumar and Urvashi Bansal

***Abstract*—A State pays for deviating from its day-ahead drawal schedule in either direction. For 34 Indian control areas we compare rules using a calibrated probability forecast of daily drawal: the forecast median, a fixed rule mixing expected regret and conditional value at risk, that rule with uncertainty- and stress-coupled risk, joint scheduling on copula scenarios and a reinforcement-learning policy. For a stylised daily model of the 2024 settlement we prove the expected regret convex, with the best risk-neutral quantile level bounded by the deviation-rate-to-price ratio. On 516 held-out days, as pre-registered, every rule beats repeating the latest recorded schedule in mean regret (fixed rule: 15.80 against 18.54 crore rupees per day), and no adaptive rule passes its test against the fixed rule. In six Eastern-Region areas the regulatory rules reproduce every official 15-minute block charge within 2.5 rupees, whereas the daily model only approximates them; re-scored per block, adaptivity again does not pay, but the forecast median beats the widened fixed rule. A better forecaster lowers mean system regret by 0.29 crore rupees daily. Schedulers should use one quantile tuned to the actual settlement and improve forecasts.**

***Index Terms*—Day-ahead scheduling, deviation settlement, conditional value at risk, scenario generation, reinforcement learning, pre-registration, Indian power system**

NOMENCLATURE

S, X	Day-ahead schedule and actual drawal of a State (GWh/day).
p, λ	Day-ahead price and normal deviation rate (Rs/MWh).
λ^+, λ^-	$\max(\lambda, p)$ and $\min(\lambda, p)$; ratio $r = \lambda/p$.
b	Tolerance band of the model, $0.1\,\lvert S\rvert$.
D^+, D^-	Over- and under-drawal, $(X-S)^+$ and $(S-X)^+$.
ℓ, L	Regret of a State and of the system (Rs crore $= 10^7$ Rs).
τ	Quantile level of a schedule.
β, ρ	Weight on $\mathrm{CVaR}_{0.9}$; scenario widening factor.
$U_{n,t}$, $Z_{n,t}$	Uncertainty, stress percentiles (State n, day t).
J	Tuning objective, $\frac{1}{2}$ mean $+$ $\frac{1}{2}$ $\mathrm{CVaR}_{0.9}$ of L.
H_{pub}	Persistence: recorded schedule of day $t-2$.
P, A1–A5c	Decision rules (Section III); R0: Section V.

R. Kumar and U. Bansal are with the Centre for Artificial Intelligence, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Jalandhar 144008, Punjab, India (e-mail: therohitkumar.build@gmail.com). ORCID: R. Kumar, 0009-0004-7936-9683; U. Bansal, 0000-0002-3395-9942. Corresponding author: R. Kumar.

I. INTRODUCTION

Indian States schedule their drawal from the inter-State grid a day ahead and settle deviations at rates set by regulation [1]. Over-drawal is charged at the normal rate, which is never below the day-ahead price, and under-drawal is credited at a fraction of it, so the cost of a deviation depends on its sign. A State with a calibrated forecast distribution of its own drawal therefore has to decide where in that distribution to place its schedule, which is a decision problem rather than a forecasting problem.

In India the National and Regional Load Despatch Centres schedule inter-State exchanges, and each State Load Despatch Centre (SLDC) is responsible for its State's drawal. Under the Indian Electricity Grid Code 2023, every SLDC must submit a day-ahead demand estimate in 15-minute time blocks by 10:00 h of the previous day, may use weather data, historical data and modern tools to prepare it, and must publish its forecast errors [2, Reg. 31(2)]. Deviations from schedule are settled per block under the Deviation Settlement Mechanism Regulations 2024, in force since 16 September 2024 [1, 3]. The normal rate is the highest of the day-ahead and real-time market clearing prices and a blend that includes the ancillary-service charge; a buyer's over-drawal is charged, and its under-drawal credited, at multiples of that rate that depend on frequency and on volume limits of 10 % of schedule or 100 MW, whichever is less [1, Regs. 7(1), 8(7)]. The system is large and changing quickly: all-India peak demand met reached 256.1 GW on 25 April 2026, with solar supplying 21.5 % of generation at that moment [4]. Indian work on the mechanism has studied the regulation and the costs it places on renewable sellers [5, 6], and an Indian SLDC forecasting study reports point forecasts [7]. We found no study that treats a State's day-ahead drawal schedule as a decision on a calibrated forecast distribution and tests decision rules out of sample against the settlement.

This paper compares decision rules for that choice on published data. Earlier work derives the bid of a generator that faces two-price imbalance settlement without a tolerance band [8, 9], or trains the forecaster for the decision [10–12]. Here a calibrated forecaster is held fixed; we derive the risk-neutral schedule under a banded, asymmetric settlement modelled on the Indian rules, and test whether making it adaptive pays. The rules range from the forecast median, through a fixed rule and

risk coupled to forecast uncertainty and system stress, to joint optimisation of all States and a learned policy. All use only information published before the decision, all are projected onto the same schedule limits, and all are scored by realised regret under a stylised daily settlement (Section III lists how it departs from the Regulations), which we then check against the official 15-minute settlement of six control areas. The evaluation uses seventeen months of data that were kept closed until every choice had been fixed in a pre-registration. The forecasts, their code and their accuracy are part of the open record [13].

The contributions are:

1) **A pre-registered comparison of scheduling rules** for 34 Indian State control areas over 516 days, scored once, with multiplicity control. Every rule has lower mean regret than carrying forward the latest recorded schedule, and none of the adaptive rules beats the fixed rule.
2) **The form of the best schedule under a banded settlement.** In the stylised model the risk-neutral schedule is a quantile of the forecast distribution; while the deviation stays inside the band, its level depends on the day only through the ratio of the deviation rate to the price, and outside the band also through the share of forecast mass beyond the band.
3) **A bound on what per-group quantile rules could earn.** Choosing the quantile level with hindsight, separately for each tercile of uncertainty, stress or price ratio, or for each State, lowers the mean regret of the best single level by at most 0.7 % under 6 settlement designs; a separate level for every day would earn 8.7 %.
4) **The chain from forecast to decision.** Holding the rule fixed and replacing the forecaster with the better one lowers mean settlement regret by about as much as the largest post hoc change of rule.
5) **A check against official 15-minute settlement.** The Regulations' buyer rules reproduce all 302,400 official block charges of six Eastern-Region areas to within Rs 2.5 (99.9 % to within one rupee); scored this way, adaptivity still does not pay, but the forecast median beats the widened fixed rule.

Contributions 2, 3 and 5 rest partly on analyses made after the confirmatory scoring; they are marked as such where they appear.

II. Related work

a) Scheduling under imbalance charges: Wherever imbalances are settled, the party that schedules faces a newsvendor-like choice on its own forecast distribution. For a wind producer paying linear imbalance penalties, the best bid is a quantile of the predictive distribution set by the ratio of the penalties [8]; this result underlies later work on trading variable generation [9], to competition between forecasters in an imbalance market [14] and to the value of probabilistic forecasts for bidding and scheduling [15]. None of these settlements has a tolerance band with a reduced credit inside it, which is the feature of the Indian rules [5, 6] that shapes the solution here. We derive the newsvendor level under such a band (Proposition 1) and test on a national system what it implies for adaptive caution.

b) Risk measures and robustness: We use CVaR [16] as the tail measure, as in risk-averse trading on combined forecasts [17], and widen the scenario set as a simple stand-in for distributional robustness.

c) Scenarios that keep dependence: Scheduling all States against a system-wide tail requires scenarios in which the States' errors move together as they actually do. Conditional joint scenarios for day-ahead planning have been generated from data with learned models [18]; we use a simpler empirical-copula construction in the spirit of the Schaake shuffle [19], in which the joint rank pattern observed on a recent published day is mapped onto today's marginal forecast quantiles, so that the dependence comes from data rather than from an assumed copula family.

d) Decisions from forecasts: That a forecast should be judged by the decision it supports is the premise of contextual optimisation and decision-focused learning [20–22]. In power systems, forecasters have been trained directly on trading cost [10], on the cost of stochastic economic dispatch [11] and on the operating cost of a system with wind power [23]; prediction intervals have been made cost-oriented [12], and the value of renewable forecast accuracy in operation has been measured directly [24]. The chain test reported here keeps the decision rule fixed and replaces the calibrated forecaster (model and calibration window), so the difference in realised regret is due to the forecast. Learned policies are the other extreme, replacing the rule as well; we include a PPO policy [25, 26] for that comparison.

e) Calibrated inputs: The distribution that feeds every arm is a stacked combination of thirteen single forecasters, calibrated by rolling conformal shifts [27]: LightGBM, XGBoost, a bidirectional LSTM, TFT, PatchTST, N-HiTS, N-BEATSx, TiDE, BiTCN, zero-shot Chronos-Bolt Small and Base and Chronos-2, and the gated LA-MCAG model. With the top-k, stacked and lag-aware online combinations they make the sixteen forecasters of Section V. Their accuracy and interval coverage are in the open record [13].

III. Risk-aware day-ahead scheduling

a) Stylised settlement: On day $t-1$ a State chooses its schedule S for day t; its actual drawal X is taken as given. With day-ahead price p, normal rate λ and a tolerance band $b = 0.1|S|$, over-drawal $D^+ = (X-S)^+$ is charged at $\lambda^+ = \max(\lambda, p)$, and at $1.2\lambda^+$ beyond the band. Under-drawal $D^- = (S-X)^+$ is credited at $0.9\lambda^-$ within the band and $0.8\lambda^-$ beyond it, where $\lambda^- = \min(\lambda, p)$. The factors 0.9 and 0.8 are the buyer's under-drawal credits at 50.00 Hz in the 2024 Regulations [1, Reg. 8(7)]. The model departs from the Regulations in five ways: (i) the Regulations credit under-drawal at a fraction of λ, and since $\lambda \geq p$ by definition [1, Reg. 7(1)] that credit can exceed p, whereas the model credits a fraction of $\min(\lambda, p) = p$; (ii) the volume limit is 10 % of the schedule or 100 MW per 15-minute block, whichever is less, which is narrower than b for a State drawing more than 1 GW on average; (iii) over-drawal beyond the limit is charged at λ or

1.5λ depending on frequency, and at 2λ with no under-drawal credit beyond a second limit, where the model uses $1.2\lambda^+$ and $0.8\lambda^-$; (iv) deviations are daily net energy, not block-wise; (v) there is no frequency linkage. The amendments in force from 23 December 2024, 1 July 2025 and 31 August 2026 change the provisions for infirm power, wind and solar sellers, storage and payment, not those for buyers [28–30]. The decision loss is the regret

$$\ell(S,X) = p\,(S-X) + \lambda^+\big[\min(D^+,b) + 1.2\,(D^+-b)^+\big] - \lambda^-\big[0.9\min(D^-,b) + 0.8\,(D^--b)^+\big] \;\geq\; 0, \quad (1)$$

which is zero only for a perfect schedule. By construction (i), no deviation earns more than an exact schedule in the model; this is a property of the model, not of the Regulations.

b) Where the best schedule sits: Because ℓ is piecewise linear in the deviation, the schedule that minimises expected regret ($\beta = 0$ below) is a quantile of the forecast distribution, as in the newsvendor problem. Write $r = \lambda/p$, $a = \max(r,1)$ and $m = \min(r,1)$. Per unit of deviation, and in units of p, the marginal regret is $a-1$ for over-drawal and $1-0.9m$ for under-drawal inside the band, and $1.2a-1$ and $1-0.8m$ beyond it. When the deviation stays inside the band, setting the derivative of the expected regret to zero gives the level of the best schedule,

$$\tau^* = \Pr(X \leq S^*) = \frac{a-1}{a-0.9m}, \quad (2)$$

which is $(r-1)/(r-0.9)$ when $r \geq 1$. That a linear imbalance penalty makes the best schedule a quantile is known from bidding under imbalance charges [8, 9]; the band and the reduced credit give the level the form (2); Proposition 1 covers deviations beyond the band.

Proposition 1: Let $S > 0$ and r, a, m be as above.
(i) Shape. For every $X > 0$, the regret $S \mapsto \ell(S,X)/p$ is convex and piecewise linear, with slopes $1-1.22a$, $1-a$, $1-0.9m$ and $1-0.81m$ on $S < X/1.1$, $X/1.1 < S < X$, $X < S < X/0.9$ and $S > X/0.9$.
(ii) Optimum. If $\Pr(X>0) = 1$ and $\mathbb{E}X < \infty$, the expected regret is convex and has a minimiser S^*, which solves

$$\begin{aligned}&(1-0.9m)\Pr(S-b \leq X < S) + (1-0.81m)\Pr(X < S-b)\\ &= (a-1)\Pr(S < X \leq S+b) + (1.22a-1)\Pr(X > S+b).\end{aligned} \quad (3)$$

(iii) Level. The level $\tau^* = \Pr(X \leq S^*)$ lies in $\big[\frac{a-1}{a-0.81m},\, \frac{1.22a-1}{1.22a-0.9m}\big]$ and equals (2) when no deviation leaves the band. For $r \geq 1$ it rises with r, with slope $0.1/(r-0.9)^2$, from 0 at $r=1$ towards 1.
Proof. (i) For $X > S$ the over-drawal is $D^+ = X - S$ and the band $b = 0.1S$ widens with S, so $\partial_S(\ell/p) = 1-a$ inside the band and $1-a[1.2(1+0.1)-0.1] = 1-1.22a$ beyond it. For $X < S$, $D^- = S-X$ and the slope is $1-0.9m$ inside the band and $1-m[0.9\cdot 0.1+0.8(1-0.1)] = 1-0.81m$ beyond it. The slopes rise with S because $1.22a \geq a \geq 1$ and $m \leq 1$, hence convexity. (ii) An expectation of convex functions is convex. Its slope tends to $1-1.22a < 0$ as $S \downarrow 0$ and to $1-0.81m > 0$ as $S \to \infty$, so a minimiser exists in $(0,\infty)$, where the slope vanishes: this is (3). (iii) Let c_o and c_u be the marginal costs per unit of over- and under-drawal at S^*, averaged over the forecast mass on each side. Then (3) reads $c_u\tau^* = c_o(1-\tau^*)$, so $\tau^* = c_o/(c_o+c_u)$, increasing in $c_o \in [a-1, 1.22a-1]$ and decreasing in $c_u \in [1-0.9m, 1-0.81m]$. The extremes give the bounds, and $c_o = a-1$, $c_u = 1-0.9m$ give (2). For $m = 1$, $\mathrm{d}\tau^*/\mathrm{d}r = 0.1/(r-0.9)^2$. □

A band that scales with the schedule keeps the loss convex but turns the single critical ratio of linear imbalance costs into a range (iii): the level depends on the forecast mass beyond the band on each side, that is on the spread relative to $0.1|S|$, not on uncertainty alone, and system stress does not enter. In development the risk weight of the fixed rule A2 was tuned to $\beta = 0$ and the widening to $\rho = 0.5$, the largest value of its grid, so A2 is the risk-neutral rule applied to a widened distribution.

c) Arms: Scenarios come from the calibrated day-ahead drawal forecast (99 interpolated quantiles). Decisions use p and λ as published at decision time (7-day means ending 14 days before the target day). $U_{n,t}$ is the calibration-year percentile of the width of the calibrated 80 % interval divided by the forecast's error scale; $Z_{n,t}$ is the mean of the percentiles of the outage share and of the share of coal capacity with less than seven days of stock (both at $t-3$) and of an indicator of energy shortage at $t-2$.

- H_{pub} (persistence): the recorded schedule of day $t-2$, the latest available at decision time, or the forecast median where it is missing. Recorded schedules include revisions made during the day, so this is not the State's own day-ahead schedule for day t.
- P: the forecast median.
- A1: a PPO policy [25, 26] that chooses the forecast quantile to schedule, with reward equal to negative regret.
- A2: minimises $(1-\beta)\,\mathbb{E}[\ell] + \beta\,\mathrm{CVaR}_{0.9}[\ell]$ [16] over scenarios whose distance from the forecast median is scaled by $(1+\rho)$, a simple robustness device.
- A4s: as A2, with the risk weight coupled to forecast uncertainty and system stress, $\beta_{n,t} = \sigma\big(c_0 + c_U(U_{n,t} - \frac{1}{2}) + c_Z(Z_{n,t} - \frac{1}{2})\big)$, σ the logistic function.
- A5: joint scheduling. All State schedules of a day are chosen together to minimise $(1-\beta)\,\mathbb{E}[L] + \beta\,\mathrm{CVaR}_{0.9}[L]$ of the *system* regret $L = \sum_n \ell(S_n, X_n)$. The joint scenarios keep the dependence between States: scenario j takes the forecast quantile ranks that all States had together on one of the 180 most recent published days and maps them onto today's forecast quantiles, an empirical-copula construction in the spirit of the Schaake shuffle [19]. The schedules for all days are found at once by gradient descent (Adam, 300 steps). With $\beta = 0$, the tuned value, the objective separates by State, and A5 differs from the risk-neutral rule only in its scenarios.
- A5c: as A5, with β for the day coupled to the mean U and mean Z across States.

Figure 1 shows where the rules sit in the pipeline. All decisions are projected onto the 1st–99th percentile range of the State's recorded schedules in the fitting period (FY2018-19 to FY2023-24) and onto the 99th percentile of their day-to-day change. These are statistical bounds, not physical limits; the States' recorded final schedules fall outside them on 20 %

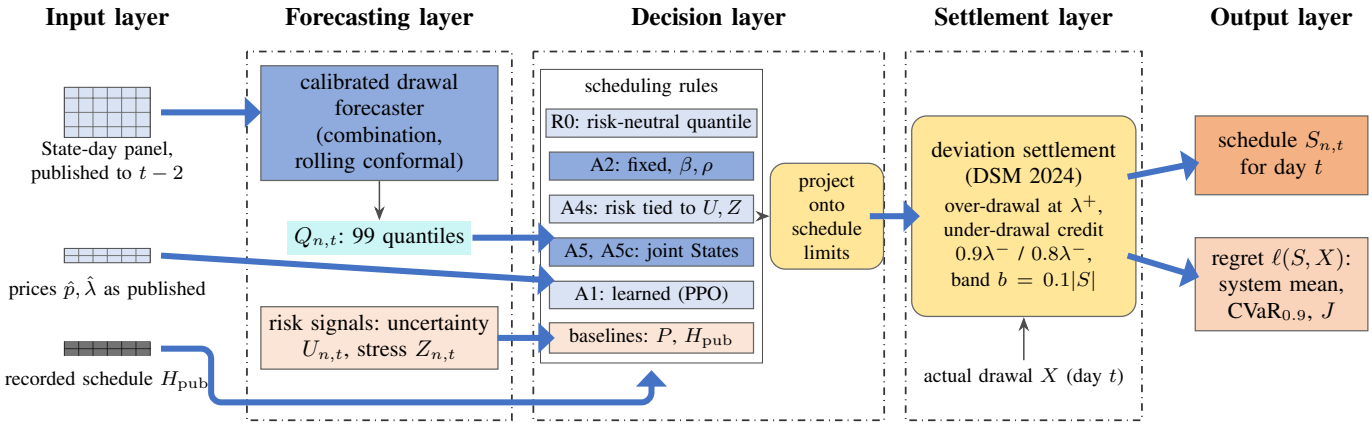


Fig. 1. Scheduling pipeline. A calibrated forecast of each State's drawal, built from data published by $t-2$, gives 99 quantiles; a scheduling rule turns them, with prices as published, into the schedule for day t; the schedule is projected onto the State's schedule limits and scored against actual drawal under the deviation settlement. All rules see the same inputs; they differ only in the decision layer.

of evaluation State-days.

IV. Data and Design

The State-day panel is built from official reports of Grid-India, the Central Electricity Authority and the India Meteorological Department [13, 31, 32]. Its 34 State control areas are all 28 States, five Union-Territory control areas and the Damodar Valley Corporation. Day-ahead drawal forecasts and their calibrated intervals come from the forecasters of Section II, fitted only on data published before each decision. In the main configuration the forecast is the combination chosen for drawal by its cross-fitted error on the calibration year (mean absolute error 5.68 GWh on the evaluation period); the first-round single model (LightGBM) serves as the comparator for the chain test.

a) Development: Development ran in two logged rounds. Weights, risk parameters, the assessment form and PPO seeds were chosen on FY2023-24, and development evidence was read on FY2024-25. Because FY2024-25 had been read once, second-round choices were made on FY2023-24 only.

b) Confirmatory evaluation: The forecasts are refitted with every boundary moved forward one year. The assessment form and its settings, and all scheduling parameters, are frozen from development; the assessment model and the PPO policies are re-fitted on the calibration year (FY2024-25) with the frozen settings. The pre-registered evaluation window is 1 April 2025 to 31 August 2026; decisions exist for 516 days, to 29 August 2026. The pre-registration, together with the hashes of the frozen choice files and of the scoring script, was committed and tagged before the evaluation data were loaded.

c) Evidence series: The main evidence series is the system-wide daily regret, summed over States. We report its mean and its $\text{CVaR}_{0.9}$ over days. For mean regret, one-sided p values come from Diebold–Mariano tests with the HLN correction [33, 34] and intervals from a 7-day moving-block bootstrap [35]. A tail day is a day on which either arm's regret is above its own 90th percentile; for tail days, p values and intervals come from a 3-day block bootstrap of the selected days. p values are Holm-adjusted within each pre-registered family [36]. The chain test compares the same arm with the combined forecaster and with the first-round forecaster, day by day.

V. Results

Numbers are from the 516 scored days of the blinded period; every parameter was frozen beforehand. Of the 9 pre-registered checks of the scheduling family, 6 are supported (Table I).

a) Against persistence and the forecast median: Every arm has lower regret than the persistence schedule. Paired differences are arm minus comparator, so a negative value favours the arm. On tail days the coupled arm differs from persistence by -7.8808 Rs crore per day (Holm-adjusted $p = 0.003$) and from the forecast median by -2.7094 ($p = 0.003$), and the joint arm differs from the median by -1.6654. In mean regret the learned policy differs from the median by -0.2814 ($p = 0.021$). Figure 2 shows that the arms are close on ordinary days and separate in the upper tail. The States' recorded final schedules, which include revisions made during the day, have a mean regret of 2.27 Rs crore per day; the day-ahead rules studied here do not compete with them. Part of the gap to persistence comes from projecting the recorded schedule onto the fixed limits: unprojected, it has mean regret 16.71, and A2 still differs from it by -0.91 ($[-1.93, -0.04]$; additional analysis).

b) Against the fixed rule: None of the adaptive arms beats the fixed rule A2. On tail days the coupled arm is slightly worse than A2 (+0.0306, not supported) and the joint arm clearly worse (+1.3183, not supported); the learned policy has a lower mean regret than A2 in the point estimate (-0.1621) but not distinguishably so ($p_{\text{Holm}} = 0.112$, not supported). The tuning itself, in Table III, says the same thing before the test does: on the calibration year the coupling coefficients of A4s went to the edges of their grid ($c_0 = -4$, $c_Z = 4$), where $\beta_{n,t} \leq 0.12$ and the rule is close to A2.

c) Why the fixed rule is hard to beat (additional analysis, not pre-registered): After the confirmatory scoring we checked (2) on the data, with the rule's own solver, the stored forecasts and nothing re-fitted. The ratio r of the deviation rate to the

TABLE I
PRE-REGISTERED CHECKS OF FAMILY F4 (SCHEDULING AND THE CHAIN), SCORED ONCE ON THE CONFIRMATORY PERIOD (6 OF 9 SUPPORTED). DECISION AS PRE-REGISTERED: SUPPORTED WHEN THE HOLM-ADJUSTED p WITHIN THE FAMILY IS BELOW 0.05 AND THE CHECK'S EXTRA RULE HOLDS; BENJAMINI–YEKUTIELI ACROSS ALL CHECKS IS REPORTED ONLY.

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H11a	A4s has lower tail-day system regret than A2	+0.0306	[-0.0233, +0.0732]	1.000	1.000	not supported
H11b	A4s has lower tail-day system regret than H_pub	-7.8808	[-10.5877, -5.6943]	0.003	0.005	supported
H11c	A4s has lower tail-day system regret than P	-2.7094	[-4.3857, -1.4029]	0.003	0.005	supported
H11d	A5 has lower tail-day system regret than A2	+1.3183	[+0.5310, +2.2780]	1.000	1.000	not supported
H11e	A5 has lower tail-day system regret than P	-1.6654	[-2.6855, -0.9652]	0.007	0.012	supported
H11f	PPO (A1) has lower mean system regret than A2	-0.1621	[-0.3447, +0.0674]	0.112	0.246	not supported
H11g	PPO (A1) has lower mean system regret than P	-0.2814	[-0.5343, -0.0621]	0.021	0.037	supported
H12	no decision arm breaks the hard limits	+0.0000	–	–	–	supported
H14	the chain with the final forecaster lowers mean system regret (arm A2) vs the round-1 forecaster	-0.2914	[-0.4522, -0.1457]	5.2×10^{-5}	7.4×10^{-5}	supported

TABLE II
SYSTEM-WIDE DAILY REGRET (RS CRORE) AND THE TUNING OBJECTIVE $J = \frac{1}{2}$ MEAN $+ \frac{1}{2}$ CVAR$_{0.9}$ ON THE CONFIRMATORY PERIOD.

Arm	Mean	CVaR$_{0.9}$	J
Final schedule, revised intra-day (reference)	2.27	4.65	3.46
Persistence, day $t-2$ (H_{pub})	18.54	47.47	33.00
Forecast median (P)	15.92	41.71	28.82
Fixed rule (A2)	15.80	38.51	27.15
Coupled risk (A4s)	15.82	38.54	27.18
Joint, copula scenarios (A5)	15.80	39.93	27.86
Joint, coupled (A5c)	15.80	39.97	27.89
Learned policy (PPO)	15.64	39.07	27.35

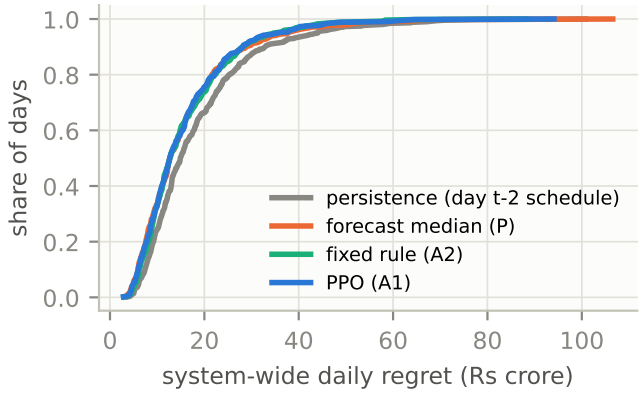


Fig. 2. Distribution of system-wide daily regret over the confirmatory period.

day-ahead price, as published before each decision, was at least 1 on 100.0 % of the 17,543 evaluation State-days, as it must be because the normal rate is defined as the highest of the market prices [1, Reg. 7(1)], with median 1.12 (5–95 %: 1.06–1.23). At the median ratio the marginal costs inside the band are $0.12\,p$ for over-drawal and $0.10\,p$ for under-drawal, close to symmetric; the fixed rule kept 64 % of its State-day deviations inside the band, so (3), not (2), governs the rest. For every State-day we solved the risk-neutral schedule and read off its level τ in the calibrated forecast distribution. The level sits in a narrow range, median 0.58 with half of all State-days between 0.52 and 0.63, and it follows r closely (rank correlation 0.87) while it barely moves with forecast uncertainty (0.06) or system stress (0.05; Fig. 3). On the calibration year the correlation with the ratio is similar (0.83) and those with uncertainty (-0.18) and stress (0.15) are weak but not zero. U measures interval width on the forecast's error scale, not relative to the band, which is what (3) makes relevant. The adaptive arms coupled their risk level to these two signals, which hardly move the optimum on this data, which explains why they gained nothing over the fixed rule.

d) How much could a per-group rule earn? (additional analysis, not pre-registered): To bound the answer we use the simplest rules: schedule at a fixed level of the calibrated distribution, chosen on a grid of five-point steps and projected onto the schedule limits like every other rule. Chosen with hindsight on the evaluation period, the best single level is 60 % (mean regret 15.45 Rs crore per day). Choosing the level separately for each tercile of forecast uncertainty, for stress on and off, or for each tercile of r lowers the mean regret by 0.00, 0.00 and 0.00 Rs crore per day (zero to two decimals), and a separate level for every State by 0.04 (0.3 %; Table IV). These are upper bounds for quantile rules whose level is constant within each group and lies on this grid; rules that vary continuously with a signal, or use other signals, are not covered. A separate level for every day gains 1.34 (8.7 %), which requires knowing the day in advance. Under five alternative settlement designs the best fixed level moves between 45 % and 70 %, but no per-group level gains more than 0.7 %. Figure 4 shows why: the regret of a fixed level is flat around its optimum, and a different settlement moves the optimum without making the curve any narrower. A change in the settlement therefore calls for re-setting the fixed level, not for an adaptive rule.

e) A rule with nothing tuned (additional analysis, not pre-registered): The risk-neutral schedule under the calibrated distribution, R0, uses no tuned parameter. On the evaluation period its mean regret is 15.63 Rs crore per day. The paired differences in daily regret, where a negative value favours the first rule, are -0.17 against A2 (interval [-0.32, +0.01]) and -0.29 against the forecast median ([-0.52, -0.11]). Its tail is heavier, with CVaR$_{0.9}$ 39.57 against 38.51 for A2, so A2 keeps the better tuning objective (J = 27.60 against 27.15). A fixed level for each State, chosen on the calibration year only, gives paired differences of -0.07 against one level for all States ([-

TABLE III
RISK PARAMETERS CARRIED INTO THE CONFIRMATORY RUN. THEY WERE CHOSEN ON THE CALIBRATION YEAR OF THE DEVELOPMENT ROUND AND THEN FROZEN; J IS THE CALIBRATION-YEAR VALUE OF THE TUNING OBJECTIVE.

Arm	Forecaster	β	ρ	c_0, c_U, c_Z	J (calibration year)
fixed risk (A2)	combined	0.00	0.50		19.48
system tail, joint scenarios (A5)	combined	0.00			20.30
coupled risk (A4s)	combined		0.50	-4.0, 0.0, 4.0	19.49
system tail, coupled (A5c)	combined			-4.0, 1.0, 2.0	20.31
fixed risk (A2)	LightGBM (chain)	0.00	0.50		20.03
system tail, joint scenarios (A5)	LightGBM (chain)	0.00			20.86

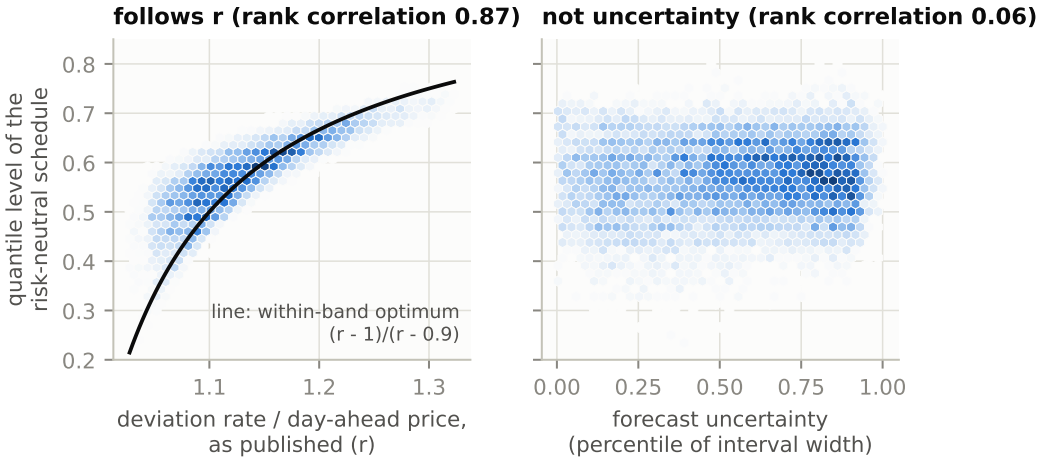


Fig. 3. Level of the risk-neutral schedule in the calibrated forecast distribution, per State-day of the evaluation period (darker cells hold more State-days), against the published rate-to-price ratio r, with the within-band optimum of (2), and against forecast uncertainty. Additional analysis, not pre-registered.

TABLE IV
LARGEST GAIN OF A QUANTILE RULE WHOSE LEVEL IS CHOSEN PER GROUP (LEVELS IN FIVE-POINT STEPS), FOUND WITH HINDSIGHT ON THE EVALUATION PERIOD: THE FALL IN MEAN SYSTEM REGRET (RS CRORE PER DAY) WHEN THE QUANTILE LEVEL IS CHOSEN PER GROUP OF A SIGNAL INSTEAD OF ONCE FOR ALL STATE-DAYS, UNDER THE STYLISED SETTLEMENT AND FIVE ALTERNATIVE DESIGNS. LEVELS IN STEPS OF FIVE POINTS. ADDITIONAL ANALYSIS, NOT PRE-REGISTERED.

Settlement design	Best level	Regret	Uncert.	Stress	r	State	Day
Stylised 2024 settlement	60 %	15.45	0.00	0.00	0.00	0.04	1.34 (8.7 %)
Narrow band (5 %)	60 %	19.09	0.01	0.00	0.01	0.06	1.43 (7.5 %)
Wide band (20 %)	60 %	12.29	0.01	0.00	0.00	0.04	1.33 (10.8 %)
Harsher over-drawal (1.5× beyond band)	70 %	19.10	0.01	0.00	0.00	0.08	1.44 (7.5 %)
Smaller under-drawal credit (0.8 / 0.6)	45 %	21.32	0.00	0.00	0.00	0.05	1.99 (9.4 %)
No credit haircut in band (1.0)	70 %	11.76	0.00	0.00	0.00	0.08	1.04 (8.9 %)

Uncert., Stress and r: terciles or states of the signal, cut on the calibration year. Day: a separate level for every day, an upper bound for any system-wide signal.

0.14, -0.01]) and -0.30 against A2 ([-0.49, -0.07]), again with a heavier tail (J = 27.41). The ranking of the rules therefore depends on the objective: for the mean regret a plain quantile rule is enough, while for the tail-weighted objective of the study the widened distribution of A2 is worth keeping.

f) How sensitive is A2 to its own settings? (additional analysis, not pre-registered): Every setting of the development grid gives J between 27.15 and 27.99 Rs crore per day on the evaluation period (Table V); the frozen setting ($\beta = 0$, $\rho = 0.5$, the edge of the grid) is also the best there.

g) A closer settlement, limits that follow demand, and learned rules (exploratory, after scoring): Four further checks use the stored forecasts and the frozen solver; none was pre-

TABLE V
TUNING OBJECTIVE J (RS CRORE PER DAY) OF THE FIXED-RISK RULE A2 ON THE EVALUATION PERIOD FOR EVERY SETTING OF ITS DEVELOPMENT GRID; BOLD: THE SETTING FROZEN IN DEVELOPMENT. ADDITIONAL ANALYSIS, NOT PRE-REGISTERED.

$\beta \setminus \rho$	0	0.1	0.25	0.5
0	27.60	27.47	27.31	**27.15**
0.25	27.69	27.58	27.45	27.49
0.5	27.77	27.68	27.58	27.74
0.75	27.78	27.72	27.66	27.86
1	27.79	27.76	27.73	27.99

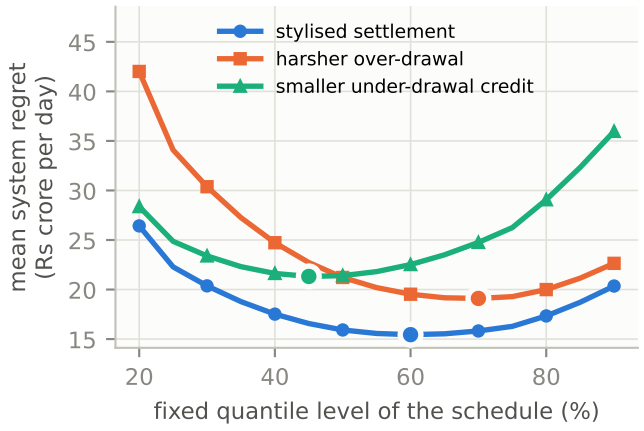


Fig. 4. Mean system regret of a fixed quantile rule against its level on the evaluation period, under the stylised settlement and two alternative designs; the larger dots mark the best level. Additional analysis, not pre-registered.

TABLE VI
CHECK AGAINST THE OFFICIAL 15-MINUTE SETTLEMENT OF SIX EASTERN-REGION AREAS (EXPLORATORY, NOT PRE-REGISTERED). TOP: DAILY CHARGE MODELS AGAINST THE OFFICIAL DAILY CHARGE; MAE AS A SHARE OF THE MEAN ABSOLUTE OFFICIAL CHARGE. BOTTOM: REGRET (RS CRORE PER DAY) WITH EVERY BLOCK SETTLED OFFICIALLY.

Daily model	Corr.	Sign (%)	MAE (%)
Stylised daily model (this paper)	0.80	85	67
Regulations, daily, at 50.00 Hz	0.81	85	71
Regulations, block frequency and rate	0.85	85	58
Rule	**Mean**	**CVaR$_{0.9}$**	***J***
Forecast median (P)	9.78	30.63	20.20
One fixed level	9.98	30.81	20.40
Adaptive level	10.04	30.91	20.48
Risk-neutral (R0)	10.22	31.02	20.62
Persistence (H_{pub})	10.41	31.80	21.11
Fixed rule (A2)	10.71	31.72	21.21

registered. First, a daily approximation of the buyer rules of the Regulations (tiers of $\min(10\,\%$ of $|S|, 100\,\text{MW})$ and $\min(15\,\%, 200\,\text{MW})$, over-drawal at λ and 2λ beyond the second tier, under-drawal credited at 0.9λ and 0.8λ and not beyond), with the second-tier rate at λ or 1.5λ (the Regulations charge 2λ beyond the second tier only below 50.00 Hz, so this approximation overstates large over-drawals), was combined with the fixed limits and with limits recomputed on the 365 days before each decision. In all six settings a level rule chosen on the calibration year per cell of r and of band-relative risk (the forecast probability that the deviation leaves the first tier) differed from one fixed level by -0.02 to +0.05 Rs crore per day, with all intervals covering zero, and no per-group level gained more than 0.46 % in hindsight; the best fixed level moved between 54 % and 57 %. Second, limits that follow demand bind on 3 % of A2's State-days and lower its regret from 15.80 to 13.49 Rs crore per day; with them A2 beats persistence by 3.0 to 13.9 Rs crore per day, projected or not, and every other rule also beats it. Third, a level policy learned directly on the settlement cost from all decision-time signals (logistic and two-layer network, protocol fixed before the evaluation run) had mean regret 15.50 against 15.49 for one fixed level chosen on the calibration year (difference +0.008, interval $[-0.036,$ +0.061]); under the closer settlement the difference was +0.011 ($[-0.122,$ +0.143]). The negative result on adaptivity therefore extends to these settlements and to continuous rules that use several signals at once. Fourth, five further baselines built from published values with nothing to fit (the same weekday's schedule a week earlier and its four-week median, the mean of the seven latest schedules, and the actual drawal of days $t-2$ and $t-7$): under all settlements and limits every one of them had higher regret than persistence, and A2 beat the best of them by 0.9 to 13.9 Rs crore per day (interval excluding zero in 11 of 12 comparisons).

h) Validation against the official 15-minute settlement (exploratory, after scoring): The Eastern Regional Power Committee publishes, for every week, the block-wise settlement of each State buyer: frequency, scheduled and actual drawal, deviation, normal rate and the amount payable or receivable [37]. For Bihar, Jharkhand, Odisha, West Bengal, Sikkim and the Damodar Valley Corporation we collected all 302,400 blocks from 31 March 2025 to 6 September 2026. The buyer rules of the Regulations, with frequency linkage and volume limits, reproduce the official charge of every block to within Rs 2.5 and of 99.9 % of blocks to within one rupee. Against the official daily charge (3,096 State-days of the evaluation period), the stylised daily model has correlation 0.80 and the right sign on 85 % of days, but a mean absolute error of 67 % of the mean absolute charge; settling the day's blocks at their own frequency and rate gives 0.85 and 58 %. The main gap is that deviations of both signs within a day are each charged, while a daily model nets them: the daily model credits these States -0.30 Rs crore per day on average, against -0.01 officially. Re-scoring the rules for the six areas with every block settled officially (each rule's daily schedule spread over the day like the State's recorded schedule; Table VI) gave mean regret of 9.78 Rs crore per day for the forecast median, 9.98 for one fixed level, 10.04 for the adaptive level, 10.22 for R0, 10.41 for persistence and 10.71 for A2. The adaptive level again did not beat the fixed level (+0.06, [+0.03, +0.10]), but the widening of A2 lost money: A2 differed from the median by +0.93 ([+0.61, +1.27]) and from persistence by +0.29 ($[-0.15,$ +0.76]).

i) Schedule limits and computation: Because every decision is projected onto the limits, no arm places a State-day outside the envelope (H12, supported); this checks the implementation rather than testing a hypothesis. The projection was active on 18 % of the State-days scheduled by A2. Every rule is cheap to run: A2 takes 0.19 ms per State-day, R0 0.16 ms, and the joint optimisation of all States and days 2.0 s in total (0.07 ms per State-day); training the five PPO seeds took 28 minutes.

j) The chain test: A better forecast does reach the decision. Holding the decision rule and every parameter fixed and replacing the calibrated forecaster (the combined forecaster with a 365-day calibration window, in place of the single model selected a year earlier with a 180-day window) gives a paired difference in mean system regret of -0.2914 Rs crore

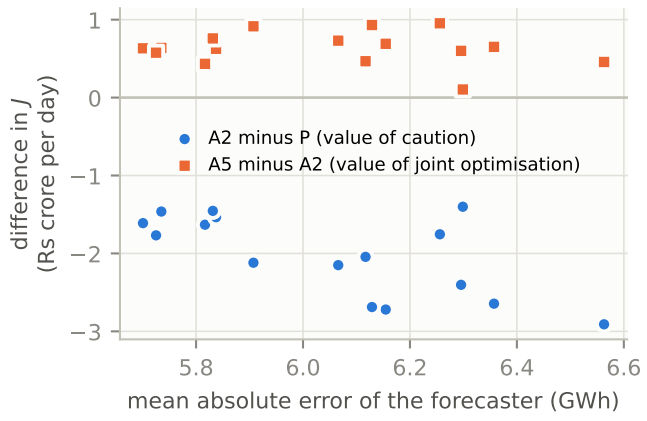


Fig. 5. Differences in the tuning objective J between rules, for each of the 16 forecasters, against the forecaster's mean absolute error; below zero favours the first rule. Additional analysis, not pre-registered.

per day (interval [-0.4522, -0.1457], $p_{\text{Holm}} = 5 \times 10^{-5}$). This is about as large as the best post hoc change of rule (R0 against P, -0.29).

k) Does the ranking depend on the forecast? (additional analysis, not pre-registered): The confirmatory comparison uses one forecaster. To see whether the ranking of the rules depends on it, we ran the frozen P, A2 and A5, unchanged, with each of the 16 forecasters (the 13 single models and 3 combinations of Section II, each with a 180-day calibration window). Their mean absolute errors range from 5.70 GWh (stacked combination) to 6.56 GWh (BiTCN). A2 has a lower tuning objective than the forecast median with 16 of the 16 forecasters, and the joint arm A5 has a lower objective than A2 with 0 of them: its mean regret is lower with 13 of them, but its tail is always heavier (Fig. 5). The gain of A2 over the median grows with the forecast error (rank correlation -0.53 between error and the difference in J). The ranking of P, A2 and A5 therefore does not depend on which of these forecasters is used.

l) Where the cost sits (descriptive, after scoring): Table VII gives the ten control areas with the largest settlement regret under the persistence schedule and what the rules would have cost instead. The saving is not concentrated in one or two States, and across all 34 control areas the fixed rule lowers mean daily regret by about a seventh relative to persistence.

m) A context-aware deviation charge (pre-registered, family F3): The pre-registration also asked whether a charge that scales size times rate by ex-ante context (renewable ramp, supply stress, carbon intensity, forecast interval width and the part of the deviation visible at scheduling time) would rank the harm of deviations better, measured by Kendall's τ_K within deciles of size times rate. On 10,759 State-days it did not (gain +0.0108, interval [-0.0154, +0.0351], not supported), although the forecast terms beat a shuffled placebo (supported, Table VIII). A sharper test needs block-level deviations and State-level frequency, which are not published.

TABLE VII
THE TEN STATES WITH THE LARGEST SETTLEMENT REGRET UNDER THE PERSISTENCE SCHEDULE, AND WHAT THE DECISION RULES WOULD HAVE COST INSTEAD (RS CRORE PER DAY). DESCRIPTIVE.

State	Persistence	Forecast median	Fixed rule (A2)	Saving (%)
TN	1.877	1.270	1.312	+30.1
PB	1.687	1.486	1.448	+14.2
HR	1.309	1.099	1.081	+17.4
UP	1.263	1.138	1.081	+14.4
GJ	1.182	1.038	1.053	+10.9
KA	1.067	0.831	0.826	+22.7
RJ	1.049	1.048	1.024	+2.4
BR	0.869	0.821	0.819	+5.7
WB	0.856	0.806	0.774	+9.6
TG	0.848	0.752	0.766	+9.6

Across all 34 control areas the fixed rule lowers mean daily regret by 14.8%.

VI. DISCUSSION

a) What the forecast distribution adds: Within the model, scheduling from a calibrated forecast distribution with any of the rules tested here has lower regret than carrying forward the latest recorded schedule. Most of that gain comes from the forecast median, and a fixed widening adds a further reduction on tail days. The States' final schedules, revised during the day, have far lower regret than any day-ahead rule, so these results concern the day-ahead step only.

b) The negative result: Adaptivity in the risk level did not pay. Coupling risk to forecast uncertainty and system stress, optimising all States jointly, and learning a policy with reinforcement learning all failed to beat the fixed rule. The settlement model explains why. Inside the band the best schedule is a quantile whose level is set by the ratio of the deviation rate to the price (2); that ratio is at least one by the definition of the normal rate and varied little, so the optimal level stayed in a narrow band, and it hardly moved with the uncertainty and stress signals the adaptive arms were built on. The calibration year pointed the same way when it pushed the coupling coefficients to the edges of their grid. Within the class of per-group quantile rules, the hindsight bounds of Table IV find no measurable gain, because the regret is flat around the optimum. The exploratory checks of Section V remove the two features that could have hidden a gain, the fixed limits and an uncertainty signal not measured relative to the band, and also try a rule learned on the settlement cost; none beat one well-chosen level.

c) Why the joint arm loses on tail days: Tuned to $\beta = 0$, the joint arm separates by State and differs from A2 mainly in having no widening; whether optimising the system tail pays is therefore not tested here.

d) What would change the answer: Under five alternative settlement designs the best fixed level moves between 45 % and 70 %, but no per-group level gains more than 0.7 %: a change in the settlement calls for re-setting the fixed level. Against the official 15-minute settlement of six Eastern-Region areas adaptivity again did not pay, but the widened fixed rule lost to the forecast median: caution tuned in a daily model does not carry over to block settlement, where deviations of both signs within a day are each charged. For an SLDC the evidence supports a short procedure: calibrate the day-

TABLE VIII
PRE-REGISTERED CHECKS OF FAMILY F3 (DEVIATION ASSESSMENT), SCORED ONCE ON THE CONFIRMATORY PERIOD (1 OF 2 SUPPORTED). DECISION AS PRE-REGISTERED: SUPPORTED WHEN THE HOLM-ADJUSTED p WITHIN THE FAMILY IS BELOW 0.05 AND THE CHECK'S EXTRA RULE HOLDS; BENJAMINI–YEKUTIELI ACROSS ALL CHECKS IS REPORTED ONLY.

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H9	assessment A ranks consequences better than M0 within M0 deciles	+0.0108	[-0.0154, +0.0351]	0.229	1.000	not supported
H10	forecast information beats the shuffled placebo (95th percentile)	+0.0287	placebo 95th pct. +0.0198	0.039	0.135	supported

ahead forecast distribution on a rolling window, schedule at one fixed quantile, set and checked against the settlement actually applied, block by block, and spend further effort on the forecast. For a regulator, (2) and (3) show how the band and the credits fix the quantile at which a cost-minimising State schedules.

e) On the learned policy: PPO was trained on the calibration year with 5 seeds and the seed best there was tested, as pre-registered. Across seeds J spans 26.78 to 27.47 Rs crore per day on the evaluation period, as much as the differences between rules: when the optimal action is nearly a constant quantile, a learned policy has little to discover and pays for the search with variance.

f) Limitations: The settlement is a daily model that departs from the Regulations in the five ways listed in Section III. The schedule limits of the confirmatory run are percentiles of past schedules, and the States' recorded final schedules fall outside them on 20 % of evaluation State-days; limits that follow demand lower every rule's regret but leave the ranking unchanged. The main baseline is persistence of the latest recorded schedule (day $t-2$), not the State's own day-ahead schedule, which is not published separately from its revisions; five further operational baselines did no better. Official block-wise settlement is public for six Eastern-Region areas only (Section V), and its re-scoring assumes that a rule's daily schedule is spread like the State's recorded schedule. Regret prices scheduled energy at the day-ahead price, while most State drawal is bought under long-term contracts. The actual drawal is taken as given. Several analyses were made after the confirmatory scoring, on a period that had already been opened, and are labelled as such. And the evaluation covers seventeen months of one system.

VII. CONCLUSION

In a pre-registered test on 516 held-out days for 34 Indian control areas, scheduling day-ahead drawal from a calibrated forecast distribution had lower regret than carrying forward the latest recorded schedule (in an exploratory check, the strongest of six operational baselines), and no adaptive rule (coupling risk to uncertainty and stress, joint optimisation, or a learned policy) beat a fixed rule. In the stylised settlement, the risk-neutral schedule is a quantile set mainly by the ratio of the deviation rate to the price, which varied little. Replacing the forecaster changed regret about as much as the best change of rule. Exploratory checks with a daily approximation of the actual buyer rules, limits that follow demand, and a rule learned on the settlement cost gave the same answer. For six Eastern-Region areas the regulatory rules reproduce all official 15-minute charges to within Rs 2.5 per block, while the daily model tracks daily charges only approximately; re-scored block by block, adaptivity again did not pay, but the forecast median beat the widened fixed rule, so the fixed level should be set for the settlement actually applied. Three extensions follow from the limits above: setting the level on block-level settlement wherever block accounts are published, using a State's own day-ahead schedule once it is published apart from its revisions, and pricing regret at the cost of the State's contract portfolio rather than at the day-ahead price.

DATA AVAILABILITY

The State-day panel is built only from reports published by Grid Controller of India, the Central Electricity Authority and the India Meteorological Department, and the original reports remain available from those agencies. The processed State-day panel, its data dictionary, the code and the frozen results are openly available on Zenodo (RE-FUSED, version 1.0.0) at https://doi.org/10.5281/zenodo.23301898, with the panel and documentation under the CC BY 4.0 licence and the code under the MIT licence. The record holds the code with the frozen results and run logs, the processed datasets, and the larger result files in four archives (parts 1 to 4, each a complete ZIP file that can be unpacked on its own), with SHA-256 checksums of every file.

CODE AVAILABILITY AND REPRODUCIBILITY

All experiments run from one driver script with fixed seeds and recorded stage logs. The pre-registration, the frozen model choices and the scoring script were hashed and tagged in the code repository before the evaluation data were opened, and every number in this paper is generated from the result files into the text as a macro. The code is released together with the panel; the DOI of the Zenodo record is https://doi.org/10.5281/zenodo.23301898. The same code, with the result files, run logs and a guide (FINAL_VERSION.md) to the code and results behind each analysis, is also openly available at https://github.com/Ohi-Me/RE-FUSED.

ACKNOWLEDGMENT

The authors acknowledge the computational facilities at Dr. B. R. Ambedkar National Institute of Technology Jalandhar, including the high-performance computing (HPC) and GPU resources used to conduct the experiments reported in this study. This research has also informed a proposal submitted

under the Central Power Research Institute (CPRI) Research Scheme on Power (RSoP) programme. Claude (Anthropic) was used in all sections of this article for English language editing, drafting of revised text, consistency checking across tables and narrative text and formatting of the revised manuscript. All experimental design, data collection, model development, the original code implementation, and interpretation of results were performed entirely by the authors, who reviewed and verified all AI-assisted output and take full responsibility for the content of this publication.

REFERENCES

[1] Central Electricity Regulatory Commission, "Central Electricity Regulatory Commission (Deviation Settlement Mechanism and Related Matters) Regulations, 2024," Notification No. L-1/260/2021/CERC, dated 5 August 2024, 2024. [Online]. Available: https://www.cercind.gov.in/regulations/192-Noti.pdf

[2] ——, "Central Electricity Regulatory Commission (Indian Electricity Grid Code) Regulations, 2023," Notification No. L-1/265/2022/CERC, dated 29 May 2023, 2023. [Online]. Available: https://www.cercind.gov.in/Regulations/180-Regulations.pdf

[3] ——, "Notification of the date of coming into force of the CERC (deviation settlement mechanism and related matters) regulations, 2024," No. L-1/260/2021/CERC, dated 27 August 2024, 2024. [Online]. Available: https://cercind.gov.in/Regulations/192-effective-160924.pdf

[4] Press Information Bureau, Ministry of Power, Government of India, "India meets all-time highest peak power demand of ∼256 GW without shortage," Release ID 2256313, 28 April 2026, 2026. [Online]. Available: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2256313

[5] S. Koul, B. Singh, and Y. S. Brar, "Improvements in deviation settlement mechanism of Indian electricity grid system through demand response management," *International Journal of Emerging Electric Power Systems*, vol. 23, no. 4, pp. 605–620, 2022.

[6] S. Mukhopadhyay and B. Bhushan, "Deviation settlement mechanism for improving grid frequency regime in India," in *2022 IEEE 10th Power India International Conference (PIICON)*, 2022, pp. 1–5.

[7] R. Bareth, M. Kochar, A. Yadav, and M. Pazoki, "Load forecasting model using LSTM for Indian state load dispatch centre," *Electrica*, vol. 24, no. 3, pp. 601–615, 2024.

[8] P. Pinson, C. Chevallier, and G. N. Kariniotakis, "Trading wind generation from short-term probabilistic forecasts of wind power," *IEEE Transactions on Power Systems*, vol. 22, no. 3, pp. 1148–1156, 2007.

[9] E. Y. Bitar, R. Rajagopal, P. P. Khargonekar, K. Poolla, and P. Varaiya, "Bringing wind energy to market," *IEEE Transactions on Power Systems*, vol. 27, no. 3, pp. 1225–1235, 2012.

[10] A. Stratigakos, S. Camal, A. Michiorri, and G. Kariniotakis, "Prescriptive trees for integrated forecasting and optimization applied in trading of renewable energy," *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4696–4708, 2022.

[11] J. Han, L. Yan, and Z. Li, "A task-based day-ahead load forecasting model for stochastic economic dispatch," *IEEE Transactions on Power Systems*, vol. 36, no. 6, pp. 5294–5304, 2021.

[12] C. Zhao, C. Wan, and Y. Song, "Cost-oriented prediction intervals: On bridging the gap between forecasting and decision," *IEEE Transactions on Power Systems*, vol. 37, no. 4, pp. 3048–3062, 2022.

[13] R. Kumar, "RE-FUSED: State-day panel of India's power system from official data, with forecasting, forecast correction and scheduling code, results and logs," Version 1.0.0, Dataset, Zenodo, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.23301898

[14] J. Cui, N. Gu, T. Zhao, C. Wu, and M. Chen, "Forecast competition in energy imbalance market," *IEEE Transactions on Power Systems*, vol. 37, no. 3, pp. 2397–2413, 2022.

[15] M. Beykirch, A. Bott, T. Janke, and F. Steinke, "The value of probabilistic forecasts for electricity market bidding and scheduling under uncertainty," *IEEE Transactions on Power Systems*, vol. 39, no. 6, pp. 6986–6997, 2024.

[16] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.

[17] J. Wang, Y. Zhou, Y. Zhang, F. Lin, and J. Wang, "Risk-averse optimal combining forecasts for renewable energy trading under CVaR assessment of forecast errors," *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 2296–2309, 2024.

[18] A. B. Krishna and A. R. Abhyankar, "An efficient data-driven conditional joint wind power scenario generation for day-ahead power system operations planning," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 3105–3117, 2024.

[19] M. Clark, S. Gangopadhyay, L. Hay, B. Rajagopalan, and R. Wilby, "The Schaake shuffle: A method for reconstructing space–time variability in forecasted precipitation and temperature fields," *Journal of Hydrometeorology*, vol. 5, no. 1, pp. 243–262, 2004.

[20] A. N. Elmachtoub and P. Grigas, "Smart "predict, then optimize"," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.

[21] U. Sadana, A. Chenreddy, E. Delage, A. Forel, E. Frejinger, and T. Vidal, "A survey of contextual optimization methods for decision-making under uncertainty," *European Journal of Operational Research*, vol. 320, no. 2, pp. 271–289, 2025.

[22] J. Mandi, J. Kotary, S. Berden, M. Mulamba, V. Bucarey, T. Guns, and F. Fioretto, "Decision-focused learning: Foundations, state of the art, benchmark and future opportunities," *Journal of Artificial Intelligence Research*, vol. 80, pp. 1623–1701, 2024.

[23] D. Wahdany, C. Schmitt, and J. L. Cremer, "More than accuracy: end-to-end wind power forecasting that optimises the energy system," *Electric Power Systems Research*, vol. 221, p. 109384, 2023.

[24] W. Yin, S. Lei, S. Feng, and Y. Hou, "Assessing the value of renewable forecasting accuracy in power system operation," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 4561–4573, 2024.

[25] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, arXiv:1707.06347.

[26] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[27] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems 34*, 2021.

[28] Central Electricity Regulatory Commission, "Central electricity regulatory commission (deviation settlement mechanism and related matters) (first amendment) regulations, 2024," 2024, notification No. L-1/260/2021/CERC, 17 December 2024, in force from 23 December 2024. [Online]. Available: https://cercind.gov.in/regulations/194-GAZ.pdf

[29] ——, "Central electricity regulatory commission (deviation settlement mechanism and related matters) (second amendment) regulations, 2025," 2025, notification No. L-1/260/2021/CERC, 25 June 2025, in force from 1 July 2025. [Online]. Available: https://cercind.gov.in/regulations/199-GAZ.pdf

[30] ——, "Central electricity regulatory commission (deviation settlement mechanism and related matters) (third amendment) regulations, 2026," 2026, notification No. L-1/260/2021/CERC, 25 August 2026, in force from 31 August 2026. [Online]. Available: https://cercind.gov.in/Regulations/208-GAZ.pdf

[31] Grid Controller of India Limited, "Daily power supply position reports, national load despatch centre," Public file interface of grid-india.in; reports 2017–2026, accessed 14 September 2026, 2026. [Online]. Available: https://grid-india.in

[32] ——, "Deviation settlement mechanism rate files, national load despatch centre," Public file interface of grid-india.in; files 2018–2026, accessed 14 September 2026, 2026. [Online]. Available: https://grid-india.in

[33] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.

[34] D. Harvey, S. Leybourne, and P. Newbold, "Testing the equality of prediction mean squared errors," *International Journal of Forecasting*, vol. 13, no. 2, pp. 281–291, 1997.

[35] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, 1989.

[36] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, pp. 65–70, 1979, jSTOR 4615733.

[37] Eastern Regional Power Committee, "DSM, SRAS, TRAS and SCUC settlement accounts, block-wise data files," 2026, weekly accounts 31 March 2025 to 6 September 2026, accessed 12 October 2026. [Online]. Available: https://www.erpc.gov.in/dsm-rras-fras-and-sced-accts